NeoSignal

AI Intelligence Daily · Multi-Source · Cross-Verified

23 October 2026

153 verified · 256 raw · 5 sources

153	4	149	0
TOTAL	VERIFIED	CONFIRMED	EMERGING

Pipeline

Scrape HackerNews (Show/Top/New), Reddit (r/artificial, r/MachineLearning, r/singularity, r/LocalLLaMA), TechCrunch AI, VentureBeat AI, MIT Tech Review, The Verge AI, Wired AI, ArXiv CS.AI

Filter 40+ AI keyword patterns — LLMs, alignment, safety, research, tooling

Dedup Title-similarity (45% threshold) merges cross-source variants

Score Authenticity = 0.5 base + 0.3 per extra source + diversity + HN score bonuses

Rank Sorted by authenticity. Below 0.25 threshold are dropped.

Authenticity Tiers

VERIFIED	**0.80 - 1.00**	3+ independent sources including media
CONFIRMED	**0.50 - 0.79**	2 sources or 1 high-quality media
EMERGING	**0.25 - 0.49**	Single-source; passes AI keyword filter

Source Breakdown

Source	Articles	Category
ArXiv CS.AI	132	Media
TechCrunch AI	12	Media
The Verge AI	6	Media
MIT Tech Review	4	Media
HackerNews	3	Community

VERIFIED INTELLIGENCE — 4 stories

VERI

Microsoft says its AI tool is better than doctors at diagnosing conditions

The Verge AI +1 source 80%

For years, AI leaders have touted healthcare as one of the clearest examples of where their technology will improve lives. Now "it is finally here," Microsoft AI CEO Mustafa Suleyman tells The Verge. The company claims to have developed an AI-powered tool it says can diagnose...

https://www.theverge.com/ai-artificial-intelligence/1016690/suleyman-microsoft-diagnost...

VERI

HarnessPAI: An Evolving Harness for Physical AI

ArXiv CS.AI +1 source 80%

arXiv:2609.29166v2 Announce Type: replace-cross Abstract: Physical AI aims to build embodied agents that perceive the world, understand and reason about it, and decide how to act. Yet the field has focused primarily on the last component: the action model that maps observations...

https://arxiv.org/abs/2609.29166

VERI

Making the case for context-aware AI

MIT Tech Review +1 source 80%

Enterprise AI can generate recommendations, automate tasks, and help organizations make decisions faster. But without an understanding of how a business actually operates, even powerful models can miss the constraints and relationships that determine whether a decision is the...

https://www.technologyreview.com/2026/10/22/1146812/making-the-case-for-context-aware-ai/

VERI

Metamodel-Guided LLM Decoding for Consistent DSL Generation

ArXiv CS.AI +1 source 80%

arXiv:2610.29749v1 Announce Type: cross Abstract: Large language models (LLMs) are increasingly used to generate software engineering artifacts written in domain-specific languages (DSLs). Correct DSL instances usually must comply with not only the concrete syntax but also the...

https://arxiv.org/abs/2610.29749

CONFIRMED SIGNALS — 149 stories

CONF

Show HN: UI micro interactions prompts for AI

HackerNews 50% HN 77

No summary — see link for full article.

https://www.typeui.sh/micro-interactions

CONF

Show HN: CATT 0.2.1: Test agents/services against a Muse/Instinct-alike, in CI

HackerNews 50% HN 7

Hey HN, we recently open sourced our Consumer Agent Testing Tool to help teams iterate against proprietary assistants like Muse and Instinct. We built this to reduce our own frustrations with building an MCP/API for these agents to consume. The problem: testing against these...

https://github.com/Coverage-Cat/consumer-agent-testing-tool

CONF

Show HN: Walkman – turn-by-turn routing in the browser, built from vector tiles

HackerNews 50% HN 3

No summary — see link for full article.

https://walkman.uz/

CONF

Uber rival inDrive eyes 2027 robotaxi tests in Central Asia with Pony.ai and Nvidia

TechCrunch AI 50%

inDrive plans to begin testing Pony.ai's robotaxis in Kazakhstan's capital in the first half of 2027.

https://techcrunch.com/2026/10/22/uber-rival-indrive-eyes-2027-robotaxi-tests-in-centra...

CONF

Left alone with nothing to do, AI agents started playing

TechCrunch AI 50%

MIT researchers marooned 13 AIs on a virtual island, and they responded by inventing games, mourning a lost ring and, eventually, getting better at their jobs.

https://techcrunch.com/2026/10/22/left-alone-with-nothing-to-do-ai-agents-started-playing/

0 CONF

OpenAI working on new $30B raise at $1.25T valuation ahead of IPO

TechCrunch AI 50%

Should OpenAI command an over $1 trillion valuation with this round, it would break records as the most highly valued private company ever.

https://techcrunch.com/2026/10/22/openai-working-on-new-30b-raise-at-1-25t-valuation-ah...

1 CONF

1Password launches a way for AI agents to log in on your behalf without seeing your password

TechCrunch AI 50%

The new feature allows agents to log into a customer's personal accounts while still protecting the private credentials that secure those accounts.

https://techcrunch.com/2026/10/22/1password-launches-a-way-for-ai-agents-to-log-in-on-y...

2 CONF

OpenAI makes its Jev competitor widely available as decision model competition heats up

TechCrunch AI 50%

OpenAI is launching its decision model to the world. Can it compete with the original Jev?

https://techcrunch.com/2026/10/22/openai-makes-its-jev-competitor-widely-available-as-d...

3 CONF

Uber puts all 30,000 employees on ChatGPT Work

TechCrunch AI 50%

The ride-share giant previously deployed a much smaller pilot for ChatGPT Work but is now taking the enterprise agent company-wide.

https://techcrunch.com/2026/10/22/uber-puts-all-30000-employees-on-chatgpt-work/

4 CONF

Spotify’s current AI bet is about being the music service that agents choose

TechCrunch AI 50%

Spotify co-CEO Gustav Söderström explains why the rise of AI agents could be good news for the music service.

https://techcrunch.com/2026/10/22/spotifys-current-ai-bet-is-about-being-the-music-serv...

5 CONF

Claude Duet app is Anthropic’s bet on agents

TechCrunch AI 50%

Claude Duet is available through waitlist for Pro and Max plan users.

https://techcrunch.com/2026/10/22/claude-duet-app-is-anthropics-bet-on-agents/

6 CONF

Nirva raises $33M for a pendant that only listens to you

TechCrunch AI 50%

Nirva will ship its $249 ambient AI pendant by the end of the year.

https://techcrunch.com/2026/10/22/nirva-raises-33m-for-a-pendant-that-only-listens-to-you/

7 CONF

DeepSeek reportedly raises $15B days after US restrictions

TechCrunch AI 50%

DeepSeek has reportedly raised $15 billion in one of China's largest funding rounds, just days after the U.S. ordered federal agencies to remove its technology over national security concerns.

https://techcrunch.com/2026/10/22/deepseek-reportedly-raises-15b-days-after-us-restrict...

8 CONF

Meta wants to pay for its broad AI buildout with a newly created class of stock

TechCrunch AI 50%

The social media giant is planning a new stock strategy to help power its broad investments in AI models and services.

https://techcrunch.com/2026/10/21/meta-wants-to-pay-for-its-broad-ai-buildout-with-a-ne...

9 CONF

The Foundations of Scalable AI

MIT Tech Review 50%

A scalable AI system requires a strong foundation. Data, engineering, platforms, operating models, and governance must converge to work as one. This session from MIT Technology Review’s EmTech Future conference explores what it takes to build that foundation in a highly...

https://www.technologyreview.com/2026/10/22/1146103/the-foundations-of-scalable-ai/

0 CONF

AI’s hidden storage bill: What inference leaves behind

MIT Tech Review 50%

Ask most executives to picture the cost of AI and they’ll likely see GPUs—racks of accelerators with gigawatts of power—and a waitlist for silicon. But that picture leaves something out: Data storage costs. What happens to data after a model responds? Much of the back-end data...

https://www.technologyreview.com/2026/10/22/1146948/ais-hidden-storage-bill-what-infere...

1 CONF

Meta’s AI could still scrape news sites, legal experts say

The Verge AI 50%

Meta has tweaked its business terms to remove language suggesting that it could freely train its AI models on news sites, like The Verge - but legal experts say these changes aren't enough to address their concerns. "This does nothing to restrict the 'associated with your...

https://www.theverge.com/tech/1017082/meta-terms-update-ai-training-concerns

2 CONF

Someone hacked the Switch 2 by using an old game and AI

The Verge AI 50%

The Nintendo Switch 2 has apparently been jailbroken, and a modder has already shown the console running a custom version of Mario Kart World featuring new playable characters like Cinnamoroll, the red Angry Bird, Peter Griffin, and Steve from Minecraft. OpenAI's Codex helped...

https://www.theverge.com/games/1016862/nintendo-switch-2-jailbreak-ai

3 CONF

Wildwood’s director says AI is ‘antithetical to everything that we are at this studio’

The Verge AI 50%

Wildwood is the most ambitious film yet from Laika, the stop-motion studio best known for the likes of Coraline and ParaNorman. It's a sprawling fantasy adventure that took five years to complete. The process involved more 230 puppets - including 36 of the main character Prue...

https://www.theverge.com/entertainment/1016647/wildwood-travis-knight-interview-ai-laik...

4 CONF

OpenAI, Anthropic, and Nvidia had record lobbying quarters

The Verge AI 50%

It was a busy summer for AI lobbyists. Anthropic, OpenAI, and Nvidia each had their highest-spending lobbying quarter between July and September, federal disclosures show. Nvidia spent the most of the group at $2.2 million, followed by Anthropic's $2.1 million and OpenAI's $1.5...

https://www.theverge.com/policy/1016585/openai-anthropic-nvidia-lobbying-records

5 CONF

A Small Model Asks Better: Fine-Tuning Query Clarification for Retrieval-Augmented Generation

ArXiv CS.AI 50%

arXiv:2610.28686v1 Announce Type: new Abstract: Query clarification is essential for retrieval-augmented generation (RAG) systems that face ambiguous user queries. We consider multi-turn grounded conversational scenarios where a user query is either ill-formed or ambiguous in...

https://arxiv.org/abs/2610.28686

6 CONF

Scaling Multilingual Pretraining of Pixel Language Models

ArXiv CS.AI 50%

arXiv:2610.29681v1 Announce Type: cross Abstract: Pixel language models process text as rendered images instead of a finite vocabulary, which gives them strong cross-lingual transfer abilities. However, not much is known about how these abilities are influenced by larger-scale...

https://arxiv.org/abs/2610.29681

7 CONF

EyeVQA: Benchmarking Ophthalmic Vision-Language Models from Recognition to Spatial Grounding

ArXiv CS.AI 50%

arXiv:2609.32352v3 Announce Type: replace-cross Abstract: Vision-language models (VLMs) have shown increasing potential for medical image understanding, yet their capabilities in ophthalmic imaging remain insufficiently characterized. Existing ophthalmic datasets are typically...

https://arxiv.org/abs/2609.32352

8 CONF

Decision language models as affordable and rapid alternatives to LLM judges of clinical generative AI

ArXiv CS.AI 50%

arXiv:2610.28826v1 Announce Type: new Abstract: Automated judges of clinical artificial intelligence (AI) outputs are necessary, but large language model (LLM) judges are costly and slow at scale. We compared Jev, a decision language model that returns structured judgments, with...

https://arxiv.org/abs/2610.28826

9 CONF

One-Slide Calibration of Pathology Foundation Models

ArXiv CS.AI 50%

arXiv:2610.08944v2 Announce Type: replace-cross Abstract: Scanner variation changes how pathology foundation models represent the same tissue. We introduce SlideRuler, which uses other regions within a slide as internal controls to predict a query region's scanner-induced shift....

https://arxiv.org/abs/2610.08944

0 CONF

To Manipulate or Not: Opinion Dynamics in LLM Interactions

ArXiv CS.AI 50%

arXiv:2610.28875v1 Announce Type: new Abstract: Large language models increasingly take part in extended conversations where they can both shape and be shaped by another party's views, yet most persuasion research measures only one side of that exchange. We introduce a...

https://arxiv.org/abs/2610.28875

1 CONF

Hierarchical Masked Diffusion Language Models

ArXiv CS.AI 50%

arXiv:2610.29255v1 Announce Type: cross Abstract: Masked diffusion language models have emerged as an alternative to autoregressive models, enabling parallel token generation. However, standard masked diffusion language models use a shared noise schedule across positions without...

https://arxiv.org/abs/2610.29255

2 CONF

GRAFT: Restoring the Missing Sampling Interface in Continuous Diffusion Language Models

ArXiv CS.AI 50%

arXiv:2610.29136v1 Announce Type: cross Abstract: Continuous diffusion language models generate a whole sequence in parallel, never drawing a single word at any intermediate step; only the final step turns embeddings into text. Models such as ELF now report autoregressive-level...

https://arxiv.org/abs/2610.29136

3 CONF

Behavior Beyond Text: Evaluating Refusal Transfer Failure in LLM Agents

ArXiv CS.AI 50%

arXiv:2610.28982v1 Announce Type: new Abstract: While large language models are increasingly deployed as tool-calling agents, their safety evaluations still focus primarily on their ability to refuse harmful requests in text. We investigate whether the refusal behavior in text...

https://arxiv.org/abs/2610.28982

4 CONF

PerMIT: Securing Persistent Memory for LLM Agents via Machine-Checked Integrity Theorems and Scalable Community Trust

ArXiv CS.AI 50%

arXiv:2610.28987v1 Announce Type: new Abstract: Persistent memory is a largely undefended surface for injection and poisoning, and the same risk limits memory sharing among agents. We present PerMIT, a memory architecture with provable integrity guarantees under an explicit...

https://arxiv.org/abs/2610.28987

5 CONF

AgentPolicyForge: From Executable Environments to Policy Benchmarks and Adaptive Guards

ArXiv CS.AI 50%

arXiv:2610.28989v1 Announce Type: new Abstract: Customized agent deployments define policies over tools, resources, and execution conditions. Providing these policies in the task agent's prompt does not reliably prevent violations. Training a guard requires examples of tool...

https://arxiv.org/abs/2610.28989

6 CONF

RuleSmith: Multi-Agent LLMs for Automated Game Balancing

ArXiv CS.AI 50%

arXiv:2602.06232v2 Announce Type: replace-cross Abstract: Game balancing is a longstanding challenge requiring repeated playtesting, expert intuition, and extensive manual tuning. We introduce RuleSmith, a framework that couples multi-agent LLM self-play with surrogate-based...

https://arxiv.org/abs/2602.06232

7 CONF

VHDLBench: Benchmarking and Fine-Tuning Large Language Models for VHDL-Based Chip Design

ArXiv CS.AI 50%

arXiv:2610.28676v1 Announce Type: cross Abstract: The rapid evolution of digital circuit complexity demands innovative approaches to hardware design automation. While Large Language Models (LLMs) have influenced software development, their application to Hardware Description...

https://arxiv.org/abs/2610.28676

8 CONF

Inheritance as an Admission Test for LLM Prompt Merge Operators

ArXiv CS.AI 50%

arXiv:2610.29023v1 Announce Type: new Abstract: Large language model (LLM) agents are often improved by iterating until a fixed verifier is done. Once that verifier saturates, the loop has nothing left to climb. We instead score agents against the current population, and a...

https://arxiv.org/abs/2610.29023

9 CONF

GRAIL: From Saliency to Semantics in Time-Series Explanation

ArXiv CS.AI 50%

arXiv:2610.29036v1 Announce Type: new Abstract: Explaining a time-series model requires both locating the evidence and extracting meaningful knowledge from it. Yet a single timestep is not a semantic unit: the meaningful patterns live across many samples, not in any one of them....

https://arxiv.org/abs/2610.29036

10 CONF

AutoFraming: Synthesizing Dynamic Frames of C Code with LLMs

ArXiv CS.AI 50%

arXiv:2610.29046v1 Announce Type: new Abstract: Large language models (LLMs) have shown considerable promise in synthesizing formal specifications from source code. However, existing approaches primarily focus on functional contracts and loop invariants, while largely neglecting...

https://arxiv.org/abs/2610.29046

11 CONF

Every Part Has Its Role: Part-of-Speech Effects in LLM Post-Training

ArXiv CS.AI 50%

arXiv:2610.29843v1 Announce Type: cross Abstract: Large language models are typically trained with next-token prediction (NTP), which rewards predicting specific tokens rather than explicitly accounting for the semantic similarities between different words or expressions. Prior...

https://arxiv.org/abs/2610.29843

2 CONF

Same Answer, Different Reasons: Trace-Level Divergence in Multilingual Large Language Models

ArXiv CS.AI 50%

arXiv:2610.29092v1 Announce Type: new Abstract: Multilingual large language models are increasingly deployed in settings where visible reasoning traces support verification, education, and safety auditing, yet standard evaluations typically measure only final-answer agreement or...

https://arxiv.org/abs/2610.29092

3 CONF

GRAM: Group-wise Rank-Aware Model Merging via Subspace Alignment

ArXiv CS.AI 50%

arXiv:2610.29110v1 Announce Type: new Abstract: Merging different models into a single model is highly desirable for multi-task deployment, particularly in regulated domains like healthcare. In such settings, models are often locally fine-tuned due to strict data privacy...

https://arxiv.org/abs/2610.29110

4 CONF

NormTune: Norm-Guided Parameter-Efficient Fine-Tuning for Reasoning-Intensive Tasks

ArXiv CS.AI 50%

arXiv:2610.29134v1 Announce Type: new Abstract: Full fine-tuning has achieved remarkable success on reasoning-intensive tasks, but it often requires substantial GPU resource consumption. Parameter-efficient fine-tuning (PEFT) provides a promising alternative for reducing this...

https://arxiv.org/abs/2610.29134

5 CONF

From Scores to Safe Sets: Exact Safety--Constrained Decoding for Medication Recommendation

ArXiv CS.AI 50%

arXiv:2610.29228v1 Announce Type: new Abstract: Recent generative medication recommendation systems ultimately produce sets of drugs, but drug--drug interaction (DDI) constraints depend on the set as a whole. A natural post-hoc approach is to first predict medications and then...

https://arxiv.org/abs/2610.29228

6 CONF

Same Probabilities, Different Advice: Gain-Loss Framing Effects in High-Stakes LLM Decision Support

ArXiv CS.AI 50%

arXiv:2610.29294v1 Announce Type: new Abstract: Large language models (LLMs) are increasingly used to support consequential decisions, yet their reliability under equivalent representations of uncertainty remains unclear. Existing work has documented prompt sensitivity,...

https://arxiv.org/abs/2610.29294

7 CONF

LLM-Based Exogenous World Modeling for Dynamic Scheduling

ArXiv CS.AI 50%

arXiv:2610.29308v1 Announce Type: new Abstract: Dynamic scheduling requires dispatching online arrivals while anticipating future jobs, as poor decisions can compound downstream costs under nonlinear tardiness objectives. We show that a pretrained large language model (LLM) can...

https://arxiv.org/abs/2610.29308

8 CONF

Generative Artificial Intelligence for Internet of Things Computing: A Systematic Survey

ArXiv CS.AI 50%

arXiv:2504.07635v2 Announce Type: replace Abstract: The integration of Generative Artificial Intelligence (GenAI) within the Internet of Things (IoT) is garnering considerable interest. This growing attention stems from the continuous evolution and widespread adoption they are...

https://arxiv.org/abs/2504.07635

9 CONF

AgentMonopoly: Behavioral Patterns and Persona Control in LLM Agents

ArXiv CS.AI 50%

arXiv:2610.29330v2 Announce Type: new Abstract: How do LLM agents' strategies differ, and how much can instructions to adopt a persona change their behavior over a game? We study these questions with AgentMonopoly, an environment where agents communicate, buy and trade...

https://arxiv.org/abs/2610.29330

0 CONF

From Prompts to Posteriors: Can LLM Agents Perform Bayesian Inference in Context?

ArXiv CS.AI 50%

arXiv:2610.29331v1 Announce Type: new Abstract: Large language models (LLMs) are increasingly applied across the scientific discovery pipeline, from synthesizing literature to generating hypotheses to informing experimental design. A unifying principle behind these tasks is the...

https://arxiv.org/abs/2610.29331

1 CONF

From Association to Decision Value: Testing Paired Instruction Dependence for VLA Corpus Selection

ArXiv CS.AI 50%

arXiv:2610.29646v1 Announce Type: new Abstract: Choosing among robot demonstration corpora is a ranking problem: an upstream diagnostic must preserve which candidates produce better downstream policies, not merely correlate with average performance. We study paired instruction...

https://arxiv.org/abs/2610.29646

2 CONF

Real PMIDs, Wrong Papers: Harness Design for Biomedical Claim-Verification Agents

ArXiv CS.AI 50%

arXiv:2610.29812v1 Announce Type: new Abstract: Literature-searching language agents are becoming a standard component of scientific assistants for the life sciences, yet there is little controlled evidence on which parts of their harness matter. We run a controlled ablation on...

https://arxiv.org/abs/2610.29812

3 CONF

RegoMender: A Multi-agent Framework for Automated Rego Policy Repair

ArXiv CS.AI 50%

arXiv:2610.29456v1 Announce Type: cross Abstract: Policy-as-Code (PaC) has emerged as a foundational paradigm in cloud-native systems. However, repairing Rego policies remains challenging due to their strict syntax, declarative semantics, and specific verification requirements....

https://arxiv.org/abs/2610.29456

4 CONF

Commit or Abstain? System One Decision Models, LLMs, and Missing Legal Authority

ArXiv CS.AI 50%

arXiv:2610.29961v1 Announce Type: new Abstract: Given the enormous number of claims to be checked, verifiable legal AI requires scalable, cost-effective models that answer the practical needs of law. System One decision models are built on large language models but, instead of...

https://arxiv.org/abs/2610.29961

5 CONF

TIDE: Tree-Induced Fine-Grained Diagnostic Evaluation from Model Behaviors

ArXiv CS.AI 50%

arXiv:2610.29985v1 Announce Type: new Abstract: Benchmark performance is the dominant standard for comparing large language models (LLMs), typically summarized by a single aggregate score, such as accuracy, pass rate, or rubric score. This aggregate score provides limited...

https://arxiv.org/abs/2610.29985

6 CONF

What Git Does Not Record: Provenance Legibility in Two AI-Assisted Research Repositories

ArXiv CS.AI 50%

arXiv:2610.28655v1 Announce Type: cross Abstract: Publishers and funders increasingly ask authors to disclose how AI tools contributed to a piece of research. In computational work the repository is where an author or an auditor would look for the answer, and we ask what it can...

https://arxiv.org/abs/2610.28655

7 CONF

CompLit: Scientific Literature Search Benchmarks Must Cover Implicit, Cumulative, and Unmet Needs

ArXiv CS.AI 50%

arXiv:2610.28658v1 Announce Type: cross Abstract: How much can we trust an agent to find the paper we need? As researchers, we search the literature from what we have read toward what we have not. We thus describe what we want in our own terms, which a paper may show only...

https://arxiv.org/abs/2610.28658

8 CONF

Beyond source attribution: Address authorship and comparability decisions in an LLM-built scholarly knowledge graph

ArXiv CS.AI 50%

arXiv:2610.28663v1 Announce Type: cross Abstract: A knowledge graph needs a stable address for its evidence even where the literature offers competing classifications and agrees on none. Yet an address chosen for retrieval readily passes for the only classification available....

https://arxiv.org/abs/2610.28663

9 CONF

VerifierBench: Benchmarking the Generation of Executable Verifiers for LLM Agents

ArXiv CS.AI 50%

arXiv:2610.28679v1 Announce Type: cross Abstract: Agents need reliable ways to validate their outputs, both when completing tasks and when filtering synthetic training data. We study verifier generation: from the instruction and environment available for solving a task, an agent...

https://arxiv.org/abs/2610.28679

0 CONF

Reinforcement learning for hardware acceleration and system-level optimization on FPGA and integrated-circuit platforms: A survey

ArXiv CS.AI 50%

arXiv:2610.28699v1 Announce Type: cross Abstract: Reinforcement learning (RL) enables adaptive and autonomous decision-making in embedded, edge, and heterogeneous computing systems. Field-programmable gate arrays (FPGAs) and integrated circuits (ICs) offer low latency, energy...

https://arxiv.org/abs/2610.28699

1 CONF

STAR-XPC: Set Transformer for Estimating X-ray Transmission in Polymer Composites

ArXiv CS.AI 50%

arXiv:2610.28709v1 Announce Type: cross Abstract: Evaluating the energy transmission ratio of an X-ray beam through a polymer composite is a challenging task in radiation shielding studies. The reference solution with Monte Carlo transport simulation is highly accurate but...

https://arxiv.org/abs/2610.28709

2 CONF

SkillEval: Learning Interpretable Ability Profiles of LLMs via Cognitive Diagnosis Models

ArXiv CS.AI 50%

arXiv:2610.28767v1 Announce Type: cross Abstract: Current evaluation of LLMs on benchmarks typically aggregates performance over individual examples into a single scalar score. This approach is overly coarse and ignores the diverse scope of skills covered by benchmark datasets....

https://arxiv.org/abs/2610.28767

3 CONF

Energy-Based Transfer for Reinforcement Learning

ArXiv CS.AI 50%

arXiv:2506.16590v2 Announce Type: replace-cross Abstract: Reinforcement learning remains sample inefficient, particularly when agents must adapt over time to related but distinct tasks, as in continual learning settings. Efficiency can be improved by transferring knowledge from...

https://arxiv.org/abs/2506.16590

4 CONF

ArcFlow: Training Flow Policies for Multiple Solver Budgets

ArXiv CS.AI 50%

arXiv:2610.28788v1 Announce Type: cross Abstract: Flow-matching policies adjust inference cost through the number of solver steps, but reinforcement learning at a single fixed solver budget can hide degradation at cheaper deployment budgets and even erase useful one-step...

https://arxiv.org/abs/2610.28788

5 CONF

CALLIOPE: Private Insights into AI Use, Without Embeddings

ArXiv CS.AI 50%

arXiv:2610.28797v1 Announce Type: cross Abstract: We introduce CALLIOPE, a top-down framework for discovering topic hierarchies in large corpora of unstructured data (e.g., LLM conversations), while supporting differential privacy (DP). Unlike traditional and even recent...

https://arxiv.org/abs/2610.28797

6 CONF

Combining Long-Term Memory and LLMs Comes at a High Cost: From Prompt Injection to Persistent Compromise

ArXiv CS.AI 50%

arXiv:2610.28810v1 Announce Type: cross Abstract: The widespread adoption of generative artificial intelligence in recent years has brought a parallel rise in prompt injection and jailbreak attacks. By embedding malicious instructions, these exploits manipulate language models...

https://arxiv.org/abs/2610.28810

7 CONF

Incentivizing Tool-Use Reinforcement Learning for Post-Arrival Emergency Care Prediction

ArXiv CS.AI 50%

arXiv:2610.21449v2 Announce Type: replace Abstract: Large language models (LLMs) can reason over both structured measurements and free-text records, which makes them attractive for emergency department (ED) prediction. However, most existing LLMs still fall short of conventional...

https://arxiv.org/abs/2610.21449

8 CONF

Where Should a Stuck Agent Spend Its Next Tokens? Causal Attribution of Inference-Time Resources

ArXiv CS.AI 50%

arXiv:2610.28834v1 Announce Type: cross Abstract: Inference-time scaling is often treated as a token-budget problem, but a failed coding agent can seek qualitatively different resources: computation, information, or feedback. We introduce Banzhaf Resource Attribution (BRA), a...

https://arxiv.org/abs/2610.28834

9 CONF

How to Use Parallel Data for Multilingual Pretraining

ArXiv CS.AI 50%

arXiv:2610.28838v1 Announce Type: cross Abstract: High-quality human-translated parallel corpora can improve cross-lingual transfer, but are typically too small for large-scale pretraining. Machine-translated web corpora help overcome this limitation, making parallel data...

https://arxiv.org/abs/2610.28838

0 CONF

When Do Gaussian Priors Fail? A Latent Geometry Investigation of RL Fine-Tuning

ArXiv CS.AI 50%

arXiv:2610.28849v1 Announce Type: cross Abstract: Latent diffusion models almost universally place a Gaussian prior on the VAE latent space. For base generation this default is unproblematic; under reinforcement-learning fine-tuning, however, it carries a structural limitation....

https://arxiv.org/abs/2610.28849

1 CONF

DualHeads: Reward-tilted few-step diffusion made simple

ArXiv CS.AI 50%

arXiv:2610.28858v1 Announce Type: cross Abstract: Distilled diffusion models generate high-quality images in a few sampling steps, but aligning them with task requirements and human preferences is difficult. Methods developed for multi-step diffusion models rely on...

https://arxiv.org/abs/2610.28858

2 CONF

Agent AI Zarz\k{a}dzaj\k{a}cy Sieci\k{a} Open RAN Zorientowana na U\.zytkownika

ArXiv CS.AI 50%

arXiv:2610.28860v1 Announce Type: cross Abstract: Jednym z najwa\.zniejszych wyzwa\'n zwi\k{a}zanych z implementacj\k{a} sieci Open RAN o architekturze zorientowanej na u\.zytkownika jest odpowiednie formowanie tzw. klastr\'ow obs{\l}uguj\k{a}cych. Problem ten mo\.ze by\'c...

https://arxiv.org/abs/2610.28860

3 CONF

A General Topological Symmetry Index for Critical Orbits In Neural Network Loss Landscapes

ArXiv CS.AI 50%

arXiv:2610.28864v1 Announce Type: cross Abstract: A systematic understanding of the global structure of neural-network loss landscapes remains challenging due to their high dimensionality. Symmetries arising from network architectures and learning objectives provide a way to...

https://arxiv.org/abs/2610.28864

4 CONF

SANE: Automated Scorer-Specific Metamorphic Testing for LLM Judges

ArXiv CS.AI 50%

arXiv:2610.28886v1 Announce Type: cross Abstract: LLM judges increasingly mediate benchmark evaluation, model selection, data curation, and training. Their decisions are useful only when each judge responds reliably to changes affecting its scoring criterion. Existing methods...

https://arxiv.org/abs/2610.28886

5 CONF

SLoSa: Small, Low-Power, and Safe Foundation Models for Robotic Task Planning

ArXiv CS.AI 50%

arXiv:2610.28955v1 Announce Type: cross Abstract: Safe robotic task planning requires planners that can reliably achieve task goals while satisfying safety constraints. Classical planners search over explicit symbolic task representations and can provide formal guarantees with...

https://arxiv.org/abs/2610.28955

6 CONF

Can Quantum Layers offer Performance and Security Advantages in Vision-Language Models?

ArXiv CS.AI 50%

arXiv:2610.28964v1 Announce Type: cross Abstract: As quantum computing moves from the noisy intermediate-scale quantum (NISQ) era toward early fault-tolerant quantum computing (EFTQC), quantum-enhanced AI is shifting from a conceptual idea to experimental research. Quantum...

https://arxiv.org/abs/2610.28964

7 CONF

Unit-Aware Differentially Private Quantum Machine Learning for Clinical and Wearable Health Data

ArXiv CS.AI 50%

arXiv:2610.28968v1 Announce Type: cross Abstract: Clinical registries and wearable sensors support early health-risk assessment, yet models trained on these records may leak sensitive information when released. Differential privacy (DP) bounds this leakage by injecting noise...

https://arxiv.org/abs/2610.28968

8 CONF

From Weather to Enhanced Weathering: Weather Foundation Models for Carbon Dioxide Removal at Operational Scale

ArXiv CS.AI 50%

arXiv:2610.28973v1 Announce Type: cross Abstract: Enhanced Rock Weathering (ERW) is a carbon removal solution with the potential to remove gigatons of CO$_2$ per year, but quantifying that removal today requires dense soil sampling and laboratory analysis. We evaluate five...

https://arxiv.org/abs/2610.28973

9 CONF

Enhancing Cross-Grade Generalization in Banana Variety Classification via Hybrid CNN-Transformers and Knowledge Self-Distillation

ArXiv CS.AI 50%

arXiv:2610.29008v1 Announce Type: cross Abstract: In recent years, automated fruit classification systems have become a more critical component in precision agriculture for efficient yield estimation, quality control, and supply chain optimization. However, they still face...

https://arxiv.org/abs/2610.29008

0 CONF

HumanCentric-Scenes-AI: A Human-Curated Multimodal Dataset of AI-Generated Human-Centric Scenes Across Surveillance, Occupational Portraits, AI-Edited Images, and Short-Form Video

ArXiv CS.AI 50%

arXiv:2610.29045v1 Announce Type: cross Abstract: We introduce HumanCentric-Scenes-AI, a human-curated dataset of 296 AI-generated stimuli that depict people across four contexts in which people encounter AI-generated media. The Surveillance subset contains 85 AI-generated...

https://arxiv.org/abs/2610.29045

1 CONF

A Large-Scale Evaluation of Role-Enhanced Zero-Shot Chain-of-Thought Prompting

ArXiv CS.AI 50%

arXiv:2610.29073v1 Announce Type: cross Abstract: This paper proposes Role-Enhanced Zero-Shot Chain-of-Thought (RE-ZS-CoT), a prompting framework that systematically integrates domain-specific expert roles to enhance reasoning in large language models (LLMs) without...

https://arxiv.org/abs/2610.29073

2 CONF

LLMSeg: Training-Free Text-Prompted Image Segmentation with B-Splines

ArXiv CS.AI 50%

arXiv:2610.29409v1 Announce Type: cross Abstract: Promptable segmentation models such as SAM 3 achieve remarkable open-vocabulary performance but require dedicated architectures, large-scale annotation engines, and task-specific training. We ask a different question: can a...

https://arxiv.org/abs/2610.29409

3 CONF

UNIVERS: Learning UNIversal Voxel-wise Embedding via Randomized Synthesis for generalizable one-shot landmark localization

ArXiv CS.AI 50%

arXiv:2610.29098v1 Announce Type: cross Abstract: One-shot landmark localization in volume scans with varying fields of view (FOVs) is challenging. While existing self-supervised learning methods have shown promising performance, they rely on contrastive learning built upon...

https://arxiv.org/abs/2610.29098

4 CONF

An Eco-Friendly Tourism Trip Planner Driven by Z-Number Uncertainty: Evidence from Baku, Dubai, Abu Dhabi, Tashkent, and Istanbul

ArXiv CS.AI 50%

arXiv:2610.29120v1 Announce Type: cross Abstract: Deployed tourism trip planners ingest travel times and ratings as exact numbers, although routing services return travel times that vary with the hour of departure and a platform rating may rest on a handful of reviews or on tens...

https://arxiv.org/abs/2610.29120

5 CONF

From AI-Free Gates to AI-First Work: How Engineering Students in India Navigate University Education and Campus Placements Without AI Guidelines

ArXiv CS.AI 50%

arXiv:2610.29140v1 Announce Type: cross Abstract: Engineering graduates in India enter a technology sector that is being reshaped by generative AI, yet the route into that sector runs through examinations and campus placement screens that largely exclude AI. Institutions in...

https://arxiv.org/abs/2610.29140

6 CONF

ORSMT: Unleashing the Potential of Satisfiability Modulo Theories for Semantic Verification and Repair in LLM-Based Optimization

ArXiv CS.AI 50%

arXiv:2610.29145v1 Announce Type: cross Abstract: Large language models increasingly automate optimization modeling from natural language problems, producing written mathematics, solver code and a returned report. Verifying one optimization problem across these artifacts...

https://arxiv.org/abs/2610.29145

7 CONF

AI-Based End-to-End Automotive Design and Software Development for SDVs

ArXiv CS.AI 50%

arXiv:2610.29160v1 Announce Type: cross Abstract: This paper explores the potential of Artificial Intelligence (AI) to enable end-to-end automotive design and software development for Software-Defined Vehicles (SDVs). Based on our contributions within the Central Car Server...

https://arxiv.org/abs/2610.29160

8 CONF

Why is AI-generated language so hard to understand?

ArXiv CS.AI 50%

arXiv:2610.29162v1 Announce Type: cross Abstract: There is a widespread sense that AI-generated text has recently become harder to understand. In this study, we investigate a set of psycholinguistic features that are known to correlate with human processing difficulty (e.g.,...

https://arxiv.org/abs/2610.29162

9 CONF

TrainBridge: Agentic NPU Training Optimization Guided by GPU Compilation Evidence

ArXiv CS.AI 50%

arXiv:2610.29171v1 Announce Type: cross Abstract: We study how to improve training efficiency on Ascend NPUs. In the NPU training programs we study, framework-level compilation optimizations such as operator fusion are only partially applied, leaving fragmented computation and...

https://arxiv.org/abs/2610.29171

0 CONF

Conditional Flow Models Convert Epistemic Uncertainty into Apparent Aleatoric Variance

ArXiv CS.AI 50%

arXiv:2610.29172v1 Announce Type: cross Abstract: Conditional Flow Models (CFMs) are increasingly being used in place of deterministic predictors across machine learning, including for tasks that were classically formulated as regression. Given their success in such tasks, we...

https://arxiv.org/abs/2610.29172

1 CONF

Correlated Failures of AI Approval Monitors

ArXiv CS.AI 50%

arXiv:2610.29178v1 Announce Type: cross Abstract: Before an AI agent takes a consequential action, such as paying, messaging or changing code, several monitors check it. Their verdicts are combined by aggregating scores or by counting approvals, whatever inputs the monitors...

https://arxiv.org/abs/2610.29178

2 CONF

Resuming Vision-Language Inference across Onboard Interruptions

ArXiv CS.AI 50%

arXiv:2610.29183v1 Announce Type: cross Abstract: Executing vision-language models (VLMs) aboard low-Earth orbit satellites enables flexible processing of observation data without first downlinking it to the ground. On commercial off-the-shelf payloads, however, a request may...

https://arxiv.org/abs/2610.29183

3 CONF

LTEM: LLM-Friendly Table-Enhanced Markdown for Table Understanding and Generation

ArXiv CS.AI 50%

arXiv:2610.29188v1 Announce Type: cross Abstract: Tables in real-world documents, including scientific papers, financial reports, and statistical summaries, frequently use merged cells, multi-level headers, and sectioned layouts. Yet the lightweight formats LLMs handle best...

https://arxiv.org/abs/2610.29188

4 CONF

LOADBench: A Large Open Analog Dataset and Benchmark for Machine Learning in IC Design

ArXiv CS.AI 50%

arXiv:2610.21591v2 Announce Type: replace Abstract: Machine learning (ML) for analog IC design is constrained by limited open data and the absence of standardized evaluation: results reported in different papers are typically obtained on bespoke, often closed, datasets and with...

https://arxiv.org/abs/2610.21591

5 CONF

Flat-RLVR: Sharpness-Aware Reinforcement Learning with Verifiable Rewards for Reasoning Capacity in LLMs

ArXiv CS.AI 50%

arXiv:2610.29281v1 Announce Type: cross Abstract: Reinforcement learning with verifiable rewards (RLVR), particularly through group-relative policy optimization (GRPO), has emerged as a central paradigm for the post-training of large language models (LLMs). While RLVR is highly...

https://arxiv.org/abs/2610.29281

6 CONF

Latent space configuration for neural network finetuning, fast ranking, and embedding dimensionality reduction

ArXiv CS.AI 50%

arXiv:2610.29295v1 Announce Type: cross Abstract: Finding K most similar elements for a query in a large gallery of neural network (NN) embeddings is essential in various areas of computer science and industrial applications of artificial intelligence. This paper proposes search...

https://arxiv.org/abs/2610.29295

7 CONF

OpenMAP-HIE: Pathology Informed Score-based Diffusion Model for Unsupervised Neonatal Hypoxic-Ischemic Encephalopathy Lesion Segmentation

ArXiv CS.AI 50%

arXiv:2610.29322v2 Announce Type: cross Abstract: Accurate localization of neonatal hypoxic--ischemic encephalopathy (HIE) lesions on apparent diffusion coefficient (ADC) images is clinically important but is limited by scarce expert annotations and substantial variation in...

https://arxiv.org/abs/2610.29322

8 CONF

ChronoSynth: Constraint-Aware Question Synthesis over Temporal Knowledge Graphs

ArXiv CS.AI 50%

arXiv:2610.29339v1 Announce Type: cross Abstract: Knowledge graph question generation (KGQG) generates natural-language questions from structured graph evidence and a target answer. When the input graph is temporal, a valid question must also preserve the answer field, time...

https://arxiv.org/abs/2610.29339

9 CONF

Memory-Centric Continual Learning for AI Agents

ArXiv CS.AI 50%

arXiv:2610.29357v1 Announce Type: cross Abstract: Autonomous agents powered by foundation models can reason, plan, and use tools, but their fixed weights and context limits hinder sustained learning from experience over long-term deployment. While external memory systems attempt...

https://arxiv.org/abs/2610.29357

00 CONF

Beyond Small Talk: Modeling Disclosure Trajectories in Human-AI Conversation

ArXiv CS.AI 50%

arXiv:2610.29413v1 Announce Type: cross Abstract: Recently, we see a rise in conversational agents as a form of social and emotional support, where disclosure will unfold over time. Yet, current evaluations of these agents typically examine isolated conversational states. In...

https://arxiv.org/abs/2610.29413

01 CONF

OmniSimulator: Aligning Small Language Models for Authentic Heterogeneous Behavior Modeling

ArXiv CS.AI 50%

arXiv:2610.29472v1 Announce Type: cross Abstract: Large language models (LLMs) have shown great potential as general-purpose user simulators for interactive systems. Prior studies have demonstrated that training on high-quality data can substantially improve model reasoning and...

https://arxiv.org/abs/2610.29472

02 CONF

Generative Modeling via Nonlinear Schr\"odinger Bridges and Branching Diffusions

ArXiv CS.AI 50%

arXiv:2610.29476v1 Announce Type: cross Abstract: We propose a new generative modeling framework grounded in forward-backward partial differential equations (PDEs). These equations arise naturally in mean-field game (MFG) theory, and are closely related to nonlinear...

https://arxiv.org/abs/2610.29476

03 CONF

Unintended Side Effects of Value Training in LLMs

ArXiv CS.AI 50%

arXiv:2610.29922v1 Announce Type: cross Abstract: As LLMs are increasingly deployed in value-sensitive contexts, aligning them with human values has emerged as a post-training objective in NLP, aimed at instilling additional values and guiding model behaviour. However, current...

https://arxiv.org/abs/2610.29922

04 CONF

Are Joint-Embedding Predictive Objectives Effective for Longitudinal EHRs?

ArXiv CS.AI 50%

arXiv:2610.29528v1 Announce Type: cross Abstract: Masked language modelling (MLM) is the dominant pretraining objective for transformer encoders applied to longitudinal electronic health records (EHRs). However, reconstructing individual events may overemphasise noisy or weakly...

https://arxiv.org/abs/2610.29528

05 CONF

Block Diffusion for Longitudinal EHR: Accuracy, Efficiency, and Representation Trade-offs

ArXiv CS.AI 50%

arXiv:2610.29533v1 Announce Type: cross Abstract: Autoregressive (AR) models have proved effective for zero-shot prediction from longitudinal electronic health records, but impose strictly sequential generation on trajectories containing temporally clustered and interdependent...

https://arxiv.org/abs/2610.29533

06 CONF

Stochastic Pruning of the DPO Computation Graph: 1.4x Faster Training and Lower Peak Memory

ArXiv CS.AI 50%

arXiv:2610.29538v1 Announce Type: cross Abstract: Direct preference optimization (DPO) incurs training costs in output scoring, sequence execution, and backpropagation, so reducing supervision alone does not necessarily save computation. We introduce a unified framework for...

https://arxiv.org/abs/2610.29538

07 CONF

Improved Training of Strong Stochastic Flow Maps

ArXiv CS.AI 50%

arXiv:2610.29605v1 Announce Type: cross Abstract: Stochastic flow maps promise few-step sampling of stochastic differential equations (SDEs) by learning to directly jump between times along the solution path, yet training these models to learn this solution map remains...

https://arxiv.org/abs/2610.29605

08 CONF

An interpretable deep learning method for early hepatocellular carcinoma diagnosis on multiphase CT

ArXiv CS.AI 50%

arXiv:2610.29616v1 Announce Type: cross Abstract: Early diagnosis of hepatocellular carcinoma is critical to preserve access to curative treatment options. The Liver Imaging Reporting and Data System (LI-RADS) provides a standardized framework for non-invasive diagnosis on...

https://arxiv.org/abs/2610.29616

09 CONF

Disentangling generalization and memorization in large language models using chess

ArXiv CS.AI 50%

arXiv:2601.16823v3 Announce Type: replace-cross Abstract: Large Language Models (LLMs) exhibit remarkable capabilities, yet it remains unclear to what extent these reflect sophisticated recall or genuine reasoning ability. We introduce chess as a controlled testbed aimed at...

https://arxiv.org/abs/2601.16823

10 CONF

Lineage-Aware Search Control for LLM-Based Evolutionary Factor Mining

ArXiv CS.AI 50%

arXiv:2610.29663v1 Announce Type: cross Abstract: Formulaic alpha factor mining searches for symbolic expressions that transform market data into predictive signals. Recent LLM-based evolutionary factor mining methods use large language models as semantic variation operators....

https://arxiv.org/abs/2610.29663

11 CONF

Conflicted Records: Typed Adversarial Contracts for Auditable LLM Output Quality Enforcement

ArXiv CS.AI 50%

arXiv:2610.29669v1 Announce Type: cross Abstract: LLM-generated records increasingly populate production information systems where they are treated as data assets subject to the same quality expectations as records from operational databases. A single such record can...

https://arxiv.org/abs/2610.29669

12 CONF

Toward Recursive Self-Improvement in Physical AI

ArXiv CS.AI 50%

arXiv:2610.29677v1 Announce Type: cross Abstract: Recursive self-improvement (RSI) describes a system that improves not only its behavior but also the process by which it improves. Recent digital agents test this idea by rewriting their own code and checking each version with...

https://arxiv.org/abs/2610.29677

13 CONF

CAVE: A Structured Credit Assignment Approach for Fragmented Visual Evidence Reasoning

ArXiv CS.AI 50%

arXiv:2605.16416v2 Announce Type: replace-cross Abstract: Vision-Language Models (VLMs) have achieved strong performance on general multimodal reasoning, yet remain challenged in integrating nonlocal visual information to support semantically underdetermined visual reasoning. We...

https://arxiv.org/abs/2605.16416

14 CONF

Where Do LLMs Learn to Recommend AI?

ArXiv CS.AI 50%

arXiv:2610.29785v1 Announce Type: cross Abstract: Asked what to study, which career to pursue, or what product to build, large language models (LLMs) often recommend AI-related options. We ask where this tendency comes from by comparing models before and after fine-tuning and by...

https://arxiv.org/abs/2610.29785

15 CONF

S-EDL: Eliciting Self-Evidence from Sequence Likelihoods for Semantic Calibration of LLMs

ArXiv CS.AI 50%

arXiv:2610.29832v1 Announce Type: cross Abstract: Calibrating large language models (LLMs) in open-ended generation is uniquely challenging because many distinct token sequences can express the same underlying meaning. Prior calibration techniques target either fixed-set...

https://arxiv.org/abs/2610.29832

16 CONF

Asking a Multimodal LLM Whether Someone Fell: Zero-Shot Screening of Low-Resolution Thermal Video in a Laboratory and an Occupied Home

ArXiv CS.AI 50%

arXiv:2610.29913v1 Announce Type: cross Abstract: Thermal cameras can watch over older people living alone without capturing faces, and multimodal large language models (LLMs) can judge a short video without task-specific training. We evaluate a production multimodal LLM, Gemini...

https://arxiv.org/abs/2610.29913

17 CONF

Privacy in Context Is Being Overlooked in the Age of General-Purpose LLMs

ArXiv CS.AI 50%

arXiv:2610.29925v1 Announce Type: cross Abstract: As large language models (LLMs) continue to grow in scope and capability, privacy concerns are on the rise. Since users increasingly interact with a single general-purpose system across many distinct contexts, there is a natural...

https://arxiv.org/abs/2610.29925

18 CONF

Your Benchmark Is an Empirical Measure Over Difficulty

ArXiv CS.AI 50%

arXiv:2610.29927v1 Announce Type: cross Abstract: A benchmark is not merely a collection of test questions; it naturally induces an empirical measure over difficulty. In this work, we develop this measure-based view of benchmark evaluation. This perspective allows us to identify...

https://arxiv.org/abs/2610.29927

19 CONF

CulturalRiddles: A Multicultural Benchmark of Riddles

ArXiv CS.AI 50%

arXiv:2610.29968v1 Announce Type: cross Abstract: As LLMs expand their language coverage, evaluating how well they understand and represent diverse cultures becomes increasingly important. However, many benchmarks remain English-centric, with limited cultural coverage and a...

https://arxiv.org/abs/2610.29968

20 CONF

FaultGene: LLM-Assisted Fault-Augmented Test Generation for Hardware-in-the-Loop Validation of Autonomous Driving Systems

ArXiv CS.AI 50%

arXiv:2610.29969v1 Announce Type: cross Abstract: Reliable validation of autonomous driving cyber-physical systems requires testing not only nominal driving behavior but also failures arising from faults in sensing and embedded computation. Physical road testing provides...

https://arxiv.org/abs/2610.29969

21 CONF

HiddenLens: A Multi-Layer LLM Framework for Detecting and Resolving Hidden Assumptions in Software Requirements

ArXiv CS.AI 50%

arXiv:2610.29996v1 Announce Type: cross Abstract: Natural-language software requirements often omit assumptions about performance, operating conditions, user behaviour, and domain constraints, affecting architecture, testing, and deployment. If undiscovered until implementation,...

https://arxiv.org/abs/2610.29996

22 CONF

Building Auditable AI Text Detectors with Sparse Autoencoders

ArXiv CS.AI 50%

arXiv:2610.30006v1 Announce Type: cross Abstract: AI-generated text detectors are increasingly deployed in consequential settings---peer review, education, admissions, journalism, and content moderation---where their outputs directly affect individuals. Yet the strongest...

https://arxiv.org/abs/2610.30006

23 CONF

Stationarity Constraints Improve Both Energies and Densities of Machine-Learned Exchange-Correlation Functionals

ArXiv CS.AI 50%

arXiv:2610.30022v1 Announce Type: cross Abstract: Empirical exchange-correlation functionals are fitted on fixed orbitals but used self-consistently, so better energies often come with worse densities. We show that the Brillouin residual of a trained functional on the Kohn-Sham...

https://arxiv.org/abs/2610.30022

24 CONF

From Reasoning to Action: Benchmarking Frontier AI Models on LIBERO-10 Robot Manipulation Tasks

ArXiv CS.AI 50%

arXiv:2610.30051v1 Announce Type: cross Abstract: Recent advances in frontier AI models have demonstrated remarkable capabilities in visual understanding, reasoning, planning, and tool use, raising the possibility of using general-purpose AI models for robot manipulation without...

https://arxiv.org/abs/2610.30051

25 CONF

Parallel AI Agent Collaboration as Graph-Structured Distributed Block Optimization

ArXiv CS.AI 50%

arXiv:2610.30074v1 Announce Type: cross Abstract: Parallel AI agents can produce useful local edits yet degrade a shared solution when their edits are combined. We model this coordination problem as discrete block optimization with an exactly evaluable objective and a known...

https://arxiv.org/abs/2610.30074

26 CONF

BatchBench: Benchmarking AI Robustness to Batch Effects and Domain Shift in Biomedical Images - A JUMP Cell Painting Leaderboard

ArXiv CS.AI 50%

arXiv:2610.30104v1 Announce Type: cross Abstract: Despite recent advances, current AI systems remain highly susceptible to domain shifts, particularly in biomedical imaging, where images are usually acquired under substantially different experimental conditions. Although recent...

https://arxiv.org/abs/2610.30104

27 CONF

Multimodal Embodied Agents: A Unified Capability-centric Perspective from Computer-Use to Robot-Use

ArXiv CS.AI 50%

arXiv:2610.30108v1 Announce Type: cross Abstract: Multimodal agents (MMAs) sustain interaction through reasoning, memory, tools, and feedback, most visibly as computer-use agents, while robotic systems (RSs) couple sensing and actuation under physical dynamics. We define...

https://arxiv.org/abs/2610.30108

28 CONF

Environment Duels: A Microcosm of LLM Co-Evolution

ArXiv CS.AI 50%

arXiv:2610.30116v1 Announce Type: cross Abstract: Through a tangle of real-world interactions, from synthetic training pipelines to data shared on the open web, frontier LLMs absorb what their predecessors and competitors create. How well does each model play its role in this...

https://arxiv.org/abs/2610.30116

29 CONF

A Theoretical Analysis of Why Masked Diffusion Models Mitigate the Reversal Curse

ArXiv CS.AI 50%

arXiv:2602.02133v3 Announce Type: replace Abstract: Autoregressive language models (ARMs) suffer from the reversal curse: after learning ''A is B,'' they often fail on the reverse query ''B is A.'' Masked diffusion language models (MDMs) exhibit this failure in a much...

https://arxiv.org/abs/2602.02133

30 CONF

2-Step Agent: How a Bayesian Decision Maker Learns from AI-Decision Support

ArXiv CS.AI 50%

arXiv:2602.21889v4 Announce Type: replace Abstract: Prognostic predictions from ML models support human decision making in several fields, including high-stakes ones such as healthcare and the judiciary. Yet, we still lack a clear understanding of how decision makers learn from...

https://arxiv.org/abs/2602.21889

31 CONF

Proposals and challenges for zero-knowledge verification of frontier AI training

ArXiv CS.AI 50%

arXiv:2606.05433v3 Announce Type: replace Abstract: Proposals for Frontier AI rules often use cumulative training compute as the main criterion for identifying high-impact models. Recent governance analyses call zero-knowledge proofs promising but currently impractical at...

https://arxiv.org/abs/2606.05433

32 CONF

Language Models Give Group-Linked Fields the Wrong Weight

ArXiv CS.AI 50%

arXiv:2608.22887v2 Announce Type: replace Abstract: Large language models are increasingly used to judge people, and their answers can depend on inputs correlated with a protected attribute such as race, even when that attribute is never shown. Interpreting this reliance...

https://arxiv.org/abs/2608.22887

33 CONF

SciExam for ENSO: Can AI Agents Build Climate Models?

ArXiv CS.AI 50%

arXiv:2610.10513v2 Announce Type: replace Abstract: Language-model agents are increasingly asked to carry out open-ended scientific research, yet their results are usually graded against a known answer, a rubric, or a language-model reviewer, none of which can tell whether a new...

https://arxiv.org/abs/2610.10513

34 CONF

When Form Changes but Logic Doesn't: Building Logic-invariant LLMs through Logical Structure

ArXiv CS.AI 50%

arXiv:2610.25402v2 Announce Type: replace Abstract: Large language models (LLMs) have achieved strong performance across diverse reasoning tasks, yet it remains unclear whether this performance reflects an understanding of underlying logical structure or a reliance on shallow...

https://arxiv.org/abs/2610.25402

35 CONF

Automatic Differentiation with Constraints in Forward and Inverse Problems

ArXiv CS.AI 50%

arXiv:2610.27888v2 Announce Type: replace Abstract: Approximate solutions of partial differential equations (PDEs) on unstructured grids can be represented on a graph. Numerical optimization involving stiff evolutionary and static PDEs often requires solving a large system of...

https://arxiv.org/abs/2610.27888

36 CONF

Allocating Indivisible Goods to Strategic Agents: Pure Nash Equilibria and Fairness

ArXiv CS.AI 50%

arXiv:2109.08644v3 Announce Type: replace-cross Abstract: We consider the problem of fairly allocating a set of indivisible goods to a set of strategic agents with additive valuation functions. We assume no monetary transfers and, therefore, a mechanism in our setting is an...

https://arxiv.org/abs/2109.08644

37 CONF

Optimized Conformal Selection: Powerful Selective Inference After Data-Dependent Score Construction

ArXiv CS.AI 50%

arXiv:2411.17983v2 Announce Type: replace-cross Abstract: We study data-driven score construction in conformal selection, which uses conformal p-values and multiple testing to select "interesting" instances with large unobserved labels from a pool of unlabeled data, while...

https://arxiv.org/abs/2411.17983

38 CONF

AnnoCaseLaw: Legal Annotation Benchmark of Civil Negligence

ArXiv CS.AI 50%

arXiv:2503.00128v2 Announce Type: replace-cross Abstract: Language models offer a way to scale costly, labour-intensive expert annotation, but their ability to reproduce expert judgements remains uncertain. This is particularly challenging in law, where annotators must...

https://arxiv.org/abs/2503.00128

39 CONF

Align, Don't Blend: Orthogonal Supervision Disentanglement As the Key to Generalizable AI-Generated Image Detection

ArXiv CS.AI 50%

arXiv:2512.06746v3 Announce Type: replace-cross Abstract: Recent advances in AI-Generated Image (AIGI) detection have largely focused on model architectures, loss design, and training strategies, while overlooking a more fundamental factor: how supervision is structured....

https://arxiv.org/abs/2512.06746

40 CONF

Algorithmic Reflectivity in Large Language Models: Methodological Study of Linguistic Structure in Mentalization

ArXiv CS.AI 50%

arXiv:2512.08945v2 Announce Type: replace-cross Abstract: Mentalization is a core human capacity involving the interpretation of one's own and others' behavior in terms of underlying mental states. Large language models (LLMs) can generate linguistically reflective discourse,...

https://arxiv.org/abs/2512.08945

41 CONF

RubRIX: Rubric-Driven Risk Mitigation in Caregiver-AI Interactions

ArXiv CS.AI 50%

arXiv:2601.13235v2 Announce Type: replace-cross Abstract: Caregivers seeking AI-mediated support express complex needs---information-seeking, emotional validation, and distress cues---that warrant careful evaluation of response safety and appropriateness. Existing AI evaluation...

https://arxiv.org/abs/2601.13235

42 CONF

Biological Context Does Not Reliably Improve LLM-Guided Experimental Optimization

ArXiv CS.AI 50%

arXiv:2605.15341v2 Announce Type: replace-cross Abstract: Autonomous labs increasingly deploy large language models (LLMs) as closedloop biology optimizers that propose a design, read the outcome, then propose again. This is partly because of an assumption that biological...

https://arxiv.org/abs/2605.15341

43 CONF

HealthMamba: An Uncertainty-aware Spatiotemporal Graph State Space Model for Effective and Reliable Healthcare Facility Visit Prediction

ArXiv CS.AI 50%

arXiv:2602.05286v4 Announce Type: replace-cross Abstract: Healthcare facility visit prediction is essential for optimizing healthcare resource allocation and informing public health policy. Despite advanced machine learning methods being employed for better prediction...

https://arxiv.org/abs/2602.05286

44 CONF

Analyzing and Improving Chain-of-Thought Monitorability Through Information Theory

ArXiv CS.AI 50%

arXiv:2602.18297v2 Announce Type: replace-cross Abstract: Chain-of-thought (CoT) monitors are LLM-based systems that analyze reasoning traces to detect when outputs may exhibit attributes of interest, such as test-hacking behavior during code generation. In this paper, we use...

https://arxiv.org/abs/2602.18297

45 CONF

Data-Free Finite-Volume Convolutional Neural Networks for Steady Compressible Euler Flow around a Circular Cylinder

ArXiv CS.AI 50%

arXiv:2603.01001v2 Announce Type: replace-cross Abstract: A data-free finite-volume convolutional neural network (FV-CNN) is formulated for steady Euler flow around a cylinder at eight Mach numbers, 2--15. Shared-face fluxes define conservative spatial balances, while automatic...

https://arxiv.org/abs/2603.01001

46 CONF

StructLens: A Structural Lens for Language Models via Maximum Spanning Trees

ArXiv CS.AI 50%

arXiv:2603.03328v3 Announce Type: replace-cross Abstract: Language exhibits inherent structures, a property that explains both language acquisition and language change. Given this characteristic, we expect language models to manifest their own internal structures as well. While...

https://arxiv.org/abs/2603.03328

47 CONF

The Stepwise Informativeness Assumption: When Do Entropy Dynamics Track Reasoning Correctness in LLMs?

ArXiv CS.AI 50%

arXiv:2604.06192v2 Announce Type: replace-cross Abstract: Recent work uses entropy-based signals at multiple representation levels to study reasoning in large language models, but the field remains largely empirical. From a theoretical perspective, reductions in model...

https://arxiv.org/abs/2604.06192

48 CONF

It Takes a Forum: Resources & Stakeholders Sustaining the Online Synthetic Non-Consensual Intimate Image Creation Ecosystem

ArXiv CS.AI 50%

arXiv:2604.12190v2 Announce Type: replace-cross Abstract: Many malicious actors responsible for disseminating synthetic non-consensual intimate imagery (SNCII) operate within internet forums to exchange resources, strategies, and generated content across multiple platforms....

https://arxiv.org/abs/2604.12190

49 CONF

ProMORNA: An End-to-End Generative Framework for Full-Length mRNA Design

ArXiv CS.AI 50%

arXiv:2605.01513v2 Announce Type: replace-cross Abstract: Designing therapeutic messenger RNA (mRNA) requires generating full-length transcripts that optimally balance intracellular stability and translation efficiency. While generative models have advanced mRNA design, existing...

https://arxiv.org/abs/2605.01513

50 CONF

Is a 9B Decision Foundation Model Needed for Symptom-to-Disease Classification? Fine-Tuning Clef-Flash Against TF-IDF Baselines

ArXiv CS.AI 50%

arXiv:2610.15029v2 Announce Type: replace-cross Abstract: Disease classification from patient-reported symptom descriptions is usually approached with lightweight discriminative models or with autoregressive large language models (LLMs), which understand language broadly but...

https://arxiv.org/abs/2610.15029

51 CONF

Model Capacity Frontier: A Development Framework for LLM-Supported Social Simulation

ArXiv CS.AI 50%

arXiv:2610.24168v2 Announce Type: replace-cross Abstract: Large language model (LLM) agents are increasingly incorporated into agent-based models (ABMs) on the premise that richer, context-sensitive human behavior can improve empirical fidelity. However, LLM-supported ABMs...

https://arxiv.org/abs/2610.24168

52 CONF

TrustAgent-RFF: Agentic Diagnostic Planning for Reliable RF Fingerprinting under Domain Shift

ArXiv CS.AI 50%

arXiv:2610.24930v2 Announce Type: replace-cross Abstract: Radio frequency fingerprinting (RFF) models can achieve accuracy under source domain conditions but become unreliable when the receiver, distance, collection day, or environment changes. Incorrect predictions in unseen...

https://arxiv.org/abs/2610.24930

53 CONF

DiaLong: Benchmarking Reminiscence in Long-Context Dialogues

ArXiv CS.AI 50%

arXiv:2610.25463v2 Announce Type: replace-cross Abstract: How well can transformer-based Large Language Models (LLMs) retain and recall memories in long contexts of dialogue? LLMs are known to forget long-range factual information in their context, yet the field lacks a...

https://arxiv.org/abs/2610.25463